

NATIONAL TECHNICAL RESEARCH ORGANISATION

SIGNAL ANALYSIS & PARAMETER EXTRACTION DOSSIER

CLASSIFICATION: SECRET // AIR-GAPPED INTERNAL // REF: 994b1017ed479632

INTERCEPT METADATA & CHAIN OF CUSTODY

Source Filename : sample_4_fsk.iq
File SHA-256 Digest : 994b1017ed479632178c36025b970c413d911db274350b4047b13afbf2c95a90
Payload File Size : 524,288 bytes (512.0 KB)
Analysis Timestamp : 2026-09-18T19:57:42Z
Operator ID : SIGINT-AIRGAP-OPERATOR
AMC Model Engine : amc_resnet18_v1.0 (Deep 1D ResNet-18 + Ensemble)

EXTRACTED RF SIGNAL PARAMETERS

Parameter Description	Estimated Value	Extraction Metric / Method
Sampling Frequency (F_s)	10.0000 MSps	Baseband Nyquist Rate
Center Frequency (F_c)	+311.679 kHz	Spectral Centroid Offset
Estimated Baud Rate	210.27 kBaud	Cyclostationary Wavelet Energy
3-dB Bandwidth	7.32 kHz	Half-Power Frequency Span
Occupied Bandwidth (99%)	3818.36 kHz	Cumulative PSD 99% Energy
Estimated SNR	21.96 dB	Fourth-Order Cumulant (C_{42})
Peak-to-Average Power (PAPR)	1.84 dB	Temporal Envelope Crest Ratio
Carrier Frequency Offset (CFO)	+712585.45 Hz	Non-Linear M-th Power Loop
Spectral Flatness (Entropy)	0.0792	Wiener Entropy Index (0.0 – 1.0)

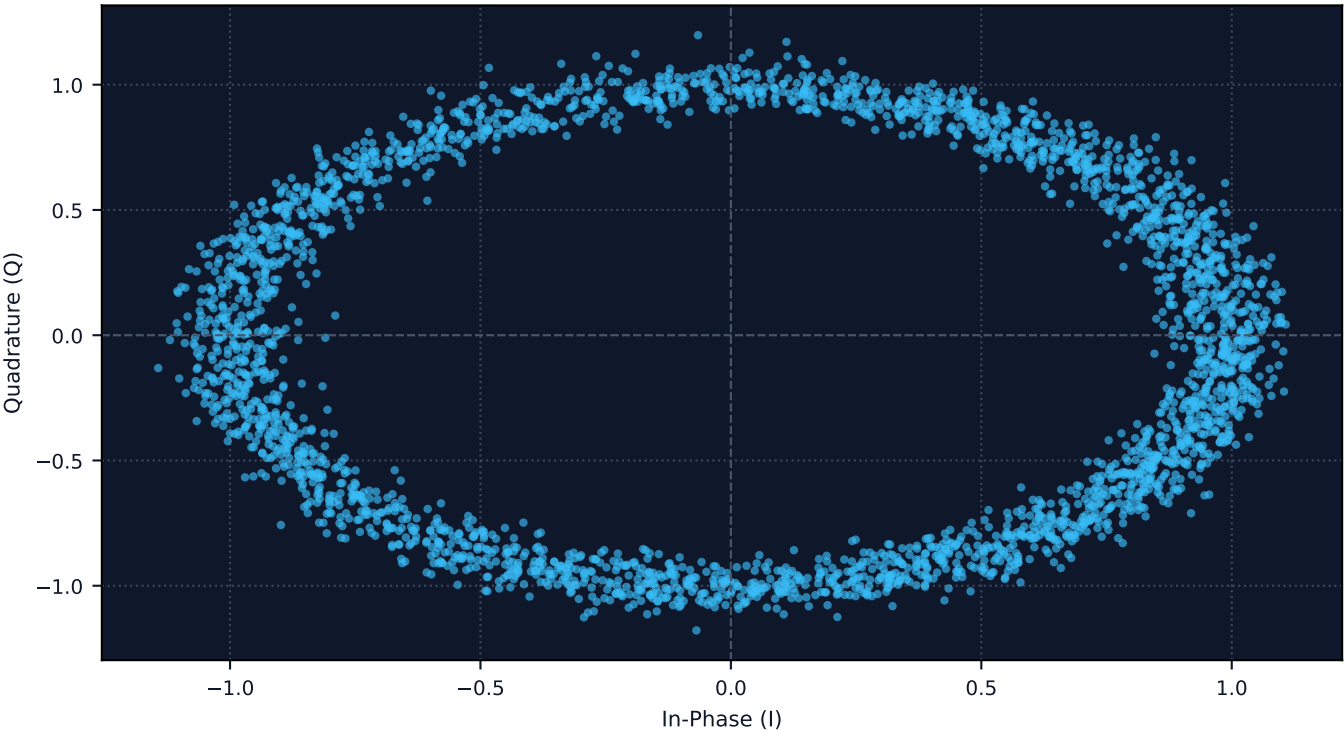
PREDICTED MODULATION : 2-FSK

CLASSIFIER CONFIDENCE : 94.00% (ResNet-18 Softmax Ensemble)

INTELLIGENCE STATUS : CONFIRMED: HIGH CERTAINTY MODULATION CLASSIFICATION

Demodulation Scheme: 2-FSK | Recovered Bits: 8,192 | Estimated EVM: 10.00% | CRC Validated Frames: 0

Demodulated Baseband Constellation Diagram — [2-FSK]



SYNCHRONIZED TELEMETRY FRAMES & PROTOCOL DISSECTION

Frame #	Sync Index	Tx ID	Seq No.	Payload	CRC Status	FEC Status
#1	292	0x0000	0	980 B	CRC MISMATCH	PARTIAL_RECOVER
#2	398	0x0000	1	967 B	CRC MISMATCH	PARTIAL_RECOVER
#3	848	0x0000	2	911 B	CRC MISMATCH	PARTIAL_RECOVER
#4	996	0x0000	3	892 B	CRC MISMATCH	PARTIAL_RECOVER
#5	1061	0x0000	4	884 B	CRC MISMATCH	PARTIAL_RECOVER

DECODED BITSTREAM HEX / ASCII PREVIEW (FIRST 64 BYTES)

0000 A7DEB18D26DF4D67FDC60226E3EC1351 [FRAME 1 PAYLOAD]